Package luahyperbolic

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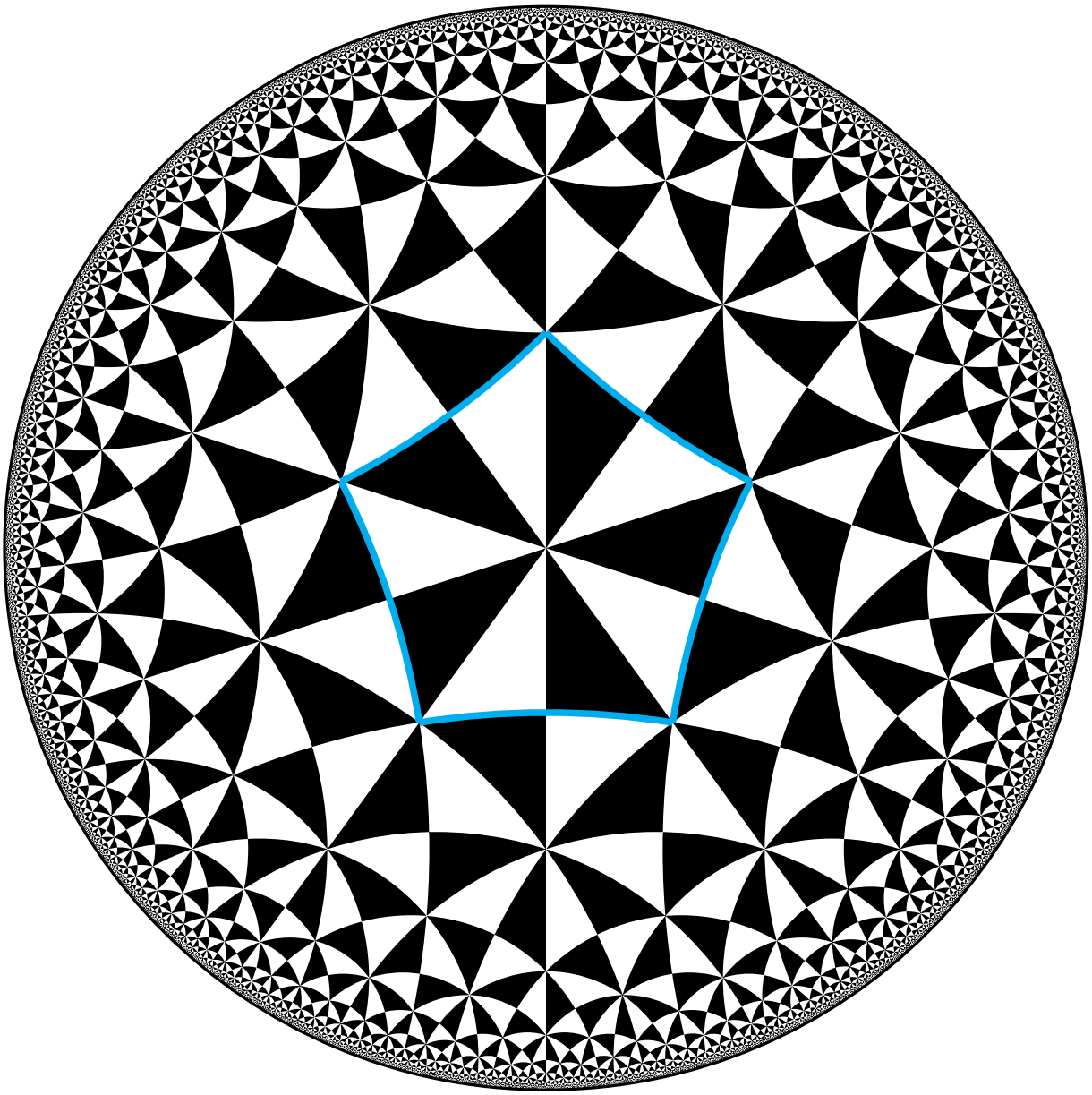

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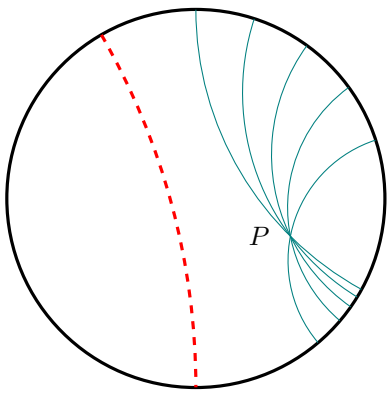
1 Introduction

1.1 What you can do with this package

1. Draw hyperbolic geometry pictures, high precision (pdf output).
2. Easy (?) syntax. At least, easier than with LaTeXmacros and pgf.
3. Easy to read, educational Lua source code if you want to learn hyperbolic geometry.
4. No need to install Python packages or create an account on some webapp.
5. No need to compile in "shell-escape" mode.
6. Compile everything in a single pass. No (Leeloo Dallas) multipass. Want to illustrate Euclid's fifth postulate ? Do it right there without leaving your document:

Draw and label points

```
\begin{hyperbolicdisk}[scale=2.5]
\begin{luacode*}
local P = complex(0.5,-0.2)
local A = complex.exp_i(math.pi/10)
for k=1,5 do
  hyper.drawLine(P, A^k, "teal")
end
hyper.labelPoint(P, "$P$", "left=.2cm")
local style = "very thick, dashed, red"
hyper.drawLine(complex.J,-complex.I,style)
\end{luacode*}
\end{hyperbolicdisk}
```



1.2 What this package cannot do for you

1. Intensive computations. Source code is not optimized to keep maximum readability. All computations involve automorphisms. For heavy computations, please use a ~~Python~~ C library. The above (embedded) image compiled in approx. 10 seconds on a 2020 intel macbook air : 1.7 seconds to compute the 3783 geodesics wit Lua, and 8 seconds to draw them with TikZ. This documentation compiles in 10s including all the tilings in last section.
2. Coffee

1.3 How to load the package

Local install

Simply put the `luahyperbolic.sty` file in your working directory and add the following line to your tex preamble :

```
\usepackage{luahyperbolic}
```

The file `luahyperbolic.sty` contains everything needed, including the necessary library for complex numbers. It can be sent to colleagues by email etc.

It will declare two lua globals : `complex` and `hyper`, and define a `hyperbolicdisk` environment for convenience.

If the sty file is not in the same directory as your tex file, you can write `\usepackage{../luahyperbolic}`

CTAN

Not available yet.

1.4 How to customize the package

The `luahyperbolic.sty` is generated from three lua modules : `complex.lua`, `luahyperbolic-core.lua` and `luahyperbolic-tikz.lua`. These files are in the `lua/` subdirectory of the package repo.

To build the `luahyperbolic.sty` from lua source files, execute `lua build.lua` in the root directory of the package. This will backup the old sty file to the `backup/` directory, then buid and replace the existing sty file.

1.5 About the code and pictures shown in this manual

In this documentation, we use the shortcut environment `\begin{hyperbolicdisk} ... \end{hyperbolicdisk}` which is defined in the package as follows:

```
\newenvironment{hyperbolicdisk}[1][scale=3]{%
  \begin{tikzpicture}[#1]
    \draw[very thick] (0,0) circle (1);
    \clip (0,0) circle (1);
}{%
  \end{tikzpicture}
}
```

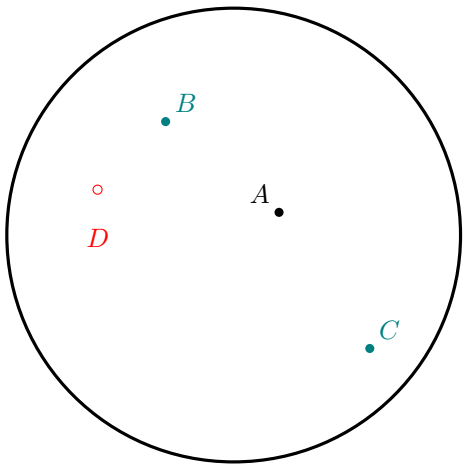
The package also provides lua functions `hyper.beginTikzPicture(options)` and `hyper.endTikzPicture(export_file)`.

2 Drawing elements

2.1 Points

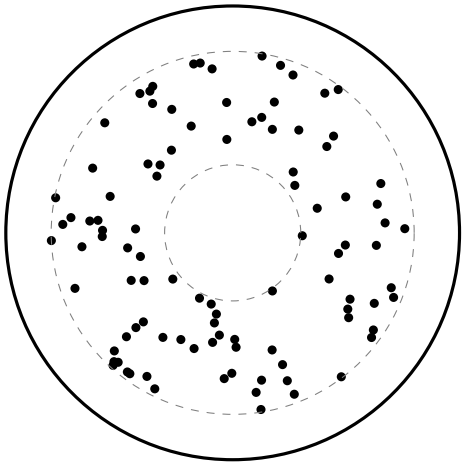
Draw and label points

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(0.2,0.1)
local B = complex(-0.3,0.5)
local C = complex(0.6,-0.5)
local D = complex(-0.6,0.2)
hyper.drawPoint(A)
hyper.drawPoints(B, C, "teal")
hyper.drawPoint(D, "white, draw=red")
hyper.labelPoint(A,"$A$") -- default : above
    ↪ left
hyper.labelPoints(B,"$B$", C, "$C$", "above
    ↪ right, teal")
hyper.labelPoint(D,"$D$","below=.4cm, red")
-- hyper.drawPoint(complex(2,2)) -- will error
\end{luacode*}
\end{hyperbolicdisk}
```



Random points with uniform density

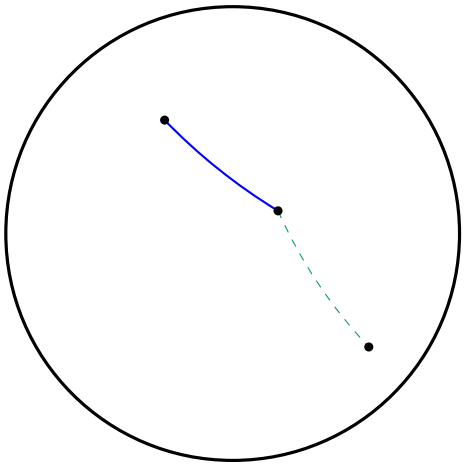
```
\begin{hyperbolicdisk}
\begin{luacode*}
for k=1,100 do
  local M = hyper.randomPoint(0.3,0.8)
  hyper.drawPoint(M)
end
\end{luacode*}
\draw[gray, dashed] (0,0) circle(0.3) circle
    ↪ (0.8);
\end{hyperbolicdisk}
```



2.2 Geodesic segments, lines and rays

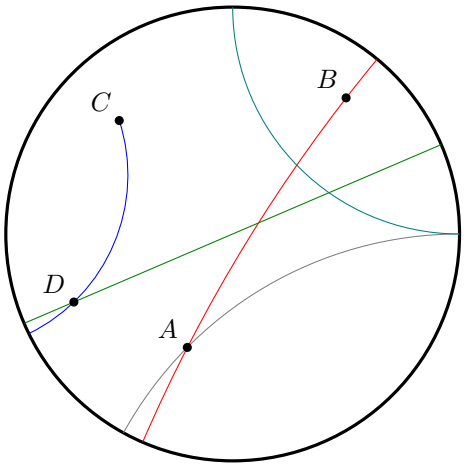
Geodesic segments

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(0.2,0.1)
local B = complex(-0.3,0.5)
local C = complex(0.6,-0.5)
hyper.drawSegment(A, B, "thick,blue")
hyper.drawSegment(A, C, "dashed,ForestGreen")
hyper.drawPoints(A,B,C)
\end{luacode*}
\end{hyperbolicdisk}
```



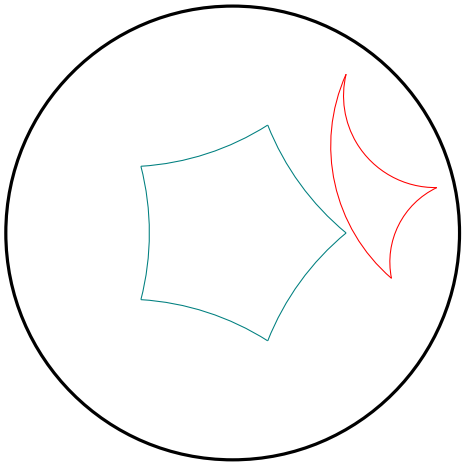
Lines and rays (through two points)

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(-0.2,-0.5)
local B = complex(0.5,0.6)
local C = complex(-0.5,0.5)
local D = complex(-0.7,-0.3)
hyper.drawLine(A, B, "red")
hyper.drawRay(C,D, "blue")
hyper.drawLine(D,-D, "green!50!black")
hyper.drawLine(A,complex(1,0), "gray")
hyper.drawLine(complex(1,0), complex(0,1), "
    ↪ teal")
hyper.drawPoints(A, B, C, D)
hyper.labelPoints(A, "$A$", B, "$B$", C, "$C$",
    ↪  D, "$D$")
\end{luacode*}
\end{hyperbolicdisk}
```



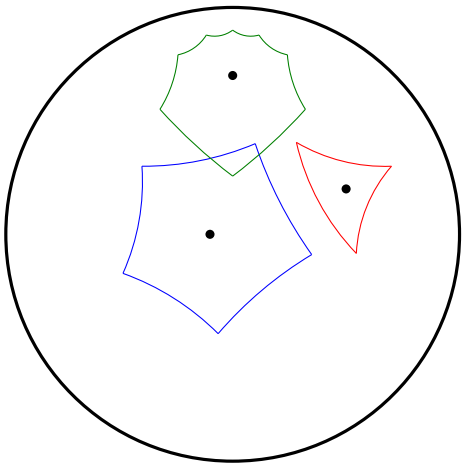
Triangles and polygons

```
\begin{hyperbolicdisk}
\begin{luacode*}
hyper.drawTriangle(
  complex(.9,.2),
  complex(.5,.7),
  complex(.7,-.2), "red")
local points={}
for k=1,5 do
  table.insert(points, complex.polar(.5,k*2*
      ↪ math.pi/5))
end
hyper.drawPolygonFromTable(points,"teal")
\end{luacode*}
\end{hyperbolicdisk}
```



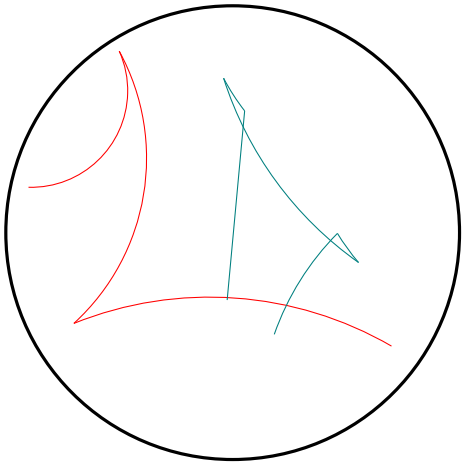
Regular polygons

```
\begin{hyperbolicdisk}
\begin{luacode*}
local O1 = complex(.5,.2)
local A1 = complex(.7,.3)
local O2 = complex(-0.1,0)
local A2 = complex(-.4,.3)
local O3 = complex(0,.7)
local A3 = complex(0,.9)
hyper.drawRegularPolygon(O1, A1, 3, "red")
hyper.drawRegularPolygon(O2, A2, 5, "blue")
hyper.drawRegularPolygon(O3, A3, 8, "green!50!
    ↪ black")
hyper.drawPoints(O1,O2,O3)
\end{luacode*}
\end{hyperbolicdisk}
```



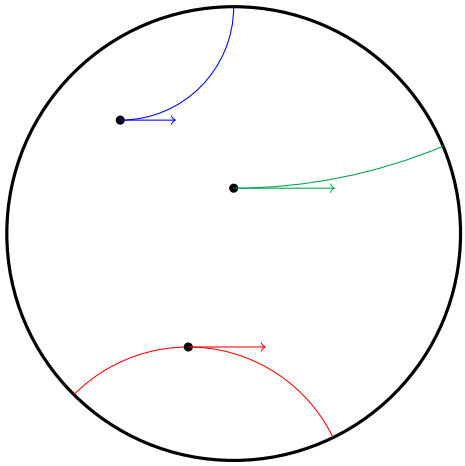
Polygonal lines

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(-.9,.2)
local B = complex(-.5,.8)
local C = complex(-.7,-.4)
local D = complex(.7,-.5)
hyper.drawPolyline(A, B, C, D, "red")
local points={}
for k=1,6 do
  table.insert(points, hyper.randomPoint(0,.7))
end
hyper.drawPolylineFromTable(points,"teal")
\end{luacode*}
\end{hyperbolicdisk}
```



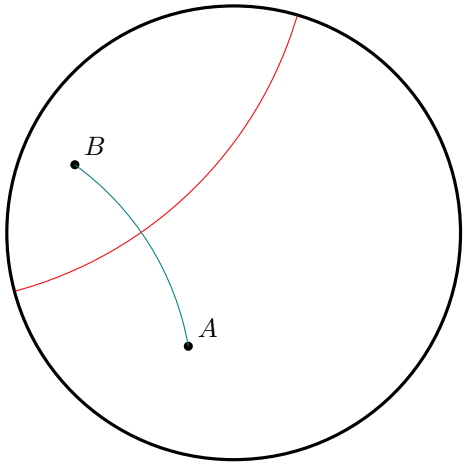
Lines and rays (point and tangent vector)

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(-0.2,-0.5)
local B = complex(-0.5,0.5)
local C = complex(0,0.2)
local v =  complex(1,0)
hyper.drawLineFromVector(A, v, "red")
hyper.drawRayFromVector(B, v, "blue")
hyper.drawRayFromVector(C, v, "Green")
hyper.drawPoints(A, B,C)
hyper.drawTangentVector(A,v,"red")
hyper.drawTangentVector(B,v,"blue")
hyper.drawTangentVector(C,v,"Green")
\end{luacode*}
\end{hyperbolicdisk}
```



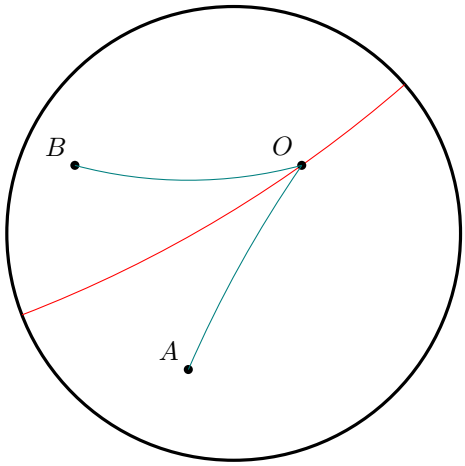
Perpendicular bisector

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(-0.2,-0.5)
local B = complex(-0.7,0.3)
hyper.drawPerpendicularBisector(A,B,"red")
hyper.drawPoints(A,B)
hyper.drawSegment(A, B, "teal")
hyper.labelPoints(A,"$A$", B, "$B$", "above
    ↪ right")
\end{luacode*}
\end{hyperbolicdisk}
```



Angle bisector

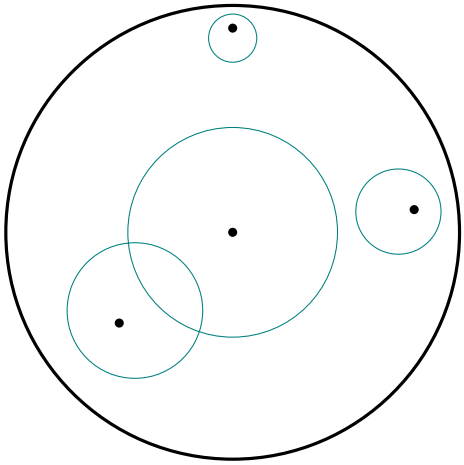
```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(-0.2,-0.6)
local O = complex(0.3,0.3)
local B = complex(-0.7,0.3)
hyper.drawAngleBisector(A,O,B,"red")
hyper.drawPoints(A,O,B)
hyper.drawPolyline(A,O,B, "teal")
hyper.labelPoints(A, "$A$", O, "$O$", B, "$B$")
\end{luacode*}
\end{hyperbolicdisk}
```



2.3 Circles, semicircles, arcs

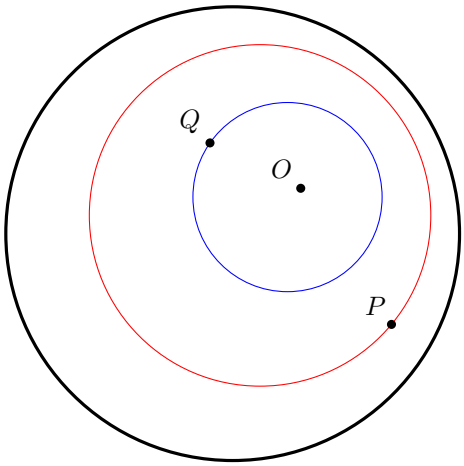
Center and radius

```
\begin{hyperbolicdisk}
\begin{luacode*}
local points = {
  complex(0,0),
  complex(-0.5,-0.4),
  complex(0.8,0.1),
  complex(0,0.9)
}
for _,point in ipairs(points) do
  hyper.drawCircle(point,1,"teal")
  hyper.drawPoint(point)
end
-- NOTE : all circles have same radius=1 !
\end{luacode*}
\end{hyperbolicdisk}
```



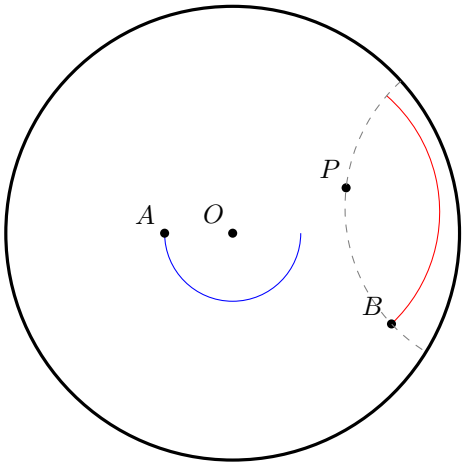
Circle through point

```
\begin{hyperbolicdisk}
\begin{luacode*}
local O = complex(0.3,0.2)
local P = complex(0.7,-0.4)
local Q = complex(-0.1,0.4)
hyper.drawCircleThrough(O, P, "red")
-- or by hand :
hyper.drawCircle(O, hyper.distance(O,Q), "blue"
    ↪ )
hyper.drawPoints(O, P, Q)
hyper.labelPoints(O, "$O$", P, "$P$", Q, "$Q$")
\end{luacode*}
\end{hyperbolicdisk}
```



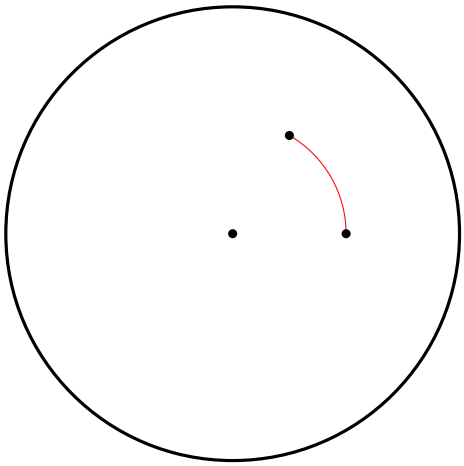
Semicircles

```
\begin{hyperbolicdisk}
\begin{luacode*}
local O = complex(0,0)
local A = complex(-0.3,0)
local P = complex(0.5,0.2)
local B = complex(0.7,-0.4)
hyper.drawSemicircle(O, A, "blue")
hyper.drawSemicircle(P, B, "red")
hyper.drawPoints(O, A, P, B)
hyper.labelPoints(O, "$O$", A, "$A$", P, "$P$",
    ↪  B, "$B$")
hyper.drawLine(P,B, "gray, dashed")
\end{luacode*}
\end{hyperbolicdisk}
```



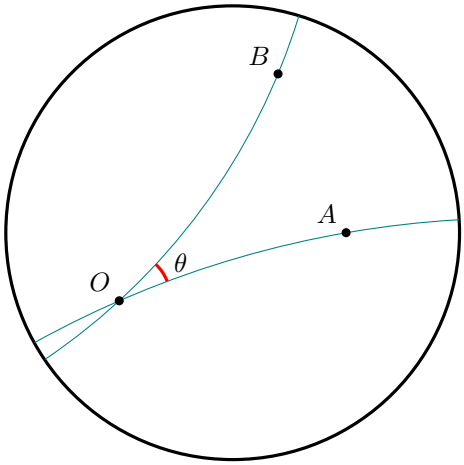
Arc (center and endpoints)

```
\begin{hyperbolicdisk}
\begin{luacode*}
local O = complex(0,0)
local A = complex(0.5,0)
local B = complex.polar(0.5,math.pi/3)
hyper.drawArc(O,A,B, "red")
hyper.drawPoints(O,A,B)
\end{luacode*}
\end{hyperbolicdisk}
```



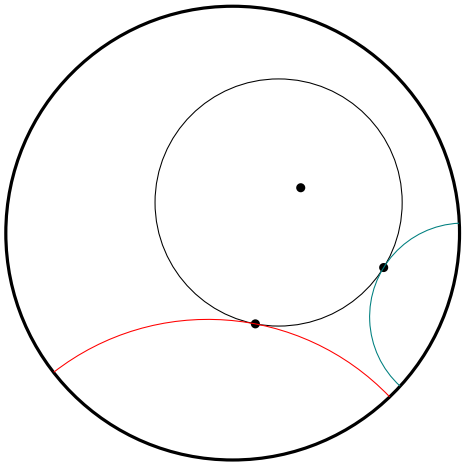
Draw angle $\widehat{AOB}$

```
\begin{hyperbolicdisk}
\begin{luacode*}
local O = complex(-.5,-.3)
local A = complex(0.5,0)
local B = complex(0.2,.7)
hyper.drawLines(O, A, O, B, "teal")
hyper.drawAngle(A,O,B, "red, very thick")
hyper.drawPoints(A,O,B)
hyper.labelPoints(A, "$A$", O, "$O$", B, "$B$")
-- poor man's labelAngle:
hyper.labelPoint(O,"$\\theta$", "above=.5cm,
    ↪ right=.6cm")
\end{luacode*}
\end{hyperbolicdisk}
```



Tangent at point

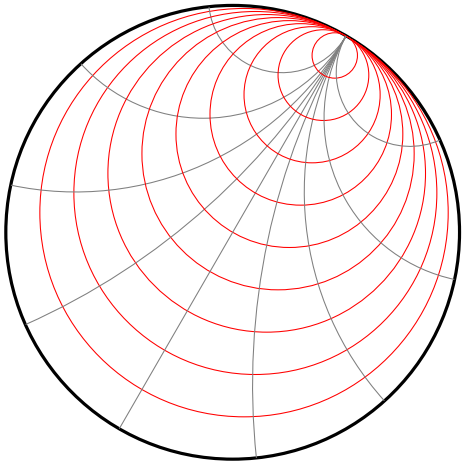
```
\begin{hyperbolicdisk}
\begin{luacode*}
local O = complex(0.3,0.2)
local P = complex(0.1,-0.4)
local Q = hyper.rotation(O,math.pi/3)(P)
hyper.drawCircleThrough(O, P)
hyper.drawPoints(O, P, Q)
hyper.drawTangentAt(O,P,"red")
hyper.drawTangentAt(O,Q,"teal")
\end{luacode*}
\end{hyperbolicdisk}
```



2.4 Horocycles

Horocycles

```
\begin{hyperbolicdisk}
\begin{luacode*}
local theta=math.pi/3
local idealPoint = complex.exp_i(theta)
local N = 10
for k=1,N-1 do
  local point = idealPoint * (2*k/N-1)
  hyper.drawHorocycle(idealPoint,point,"red")
  -- and an orthogonal geodesic :
  local endPoint = complex.exp_i(theta+2*k*math
      ↪ .pi/N)
  hyper.drawLine(idealPoint,endPoint,"gray")
end
\end{luacode*}
\end{hyperbolicdisk}
```

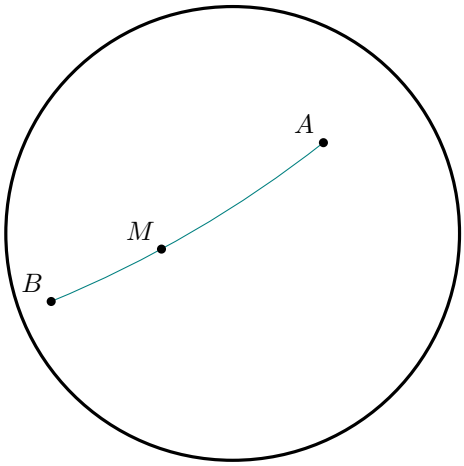


3 Defining points

3.1 Midpoint, interpolate, barycenter

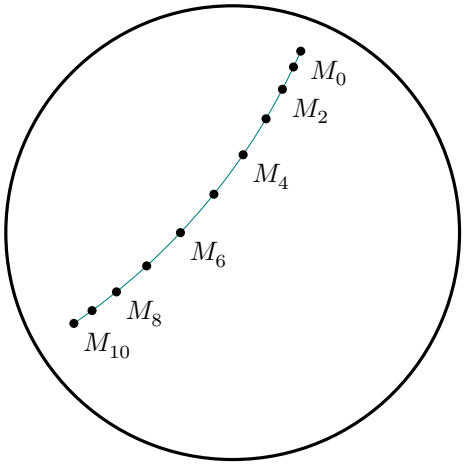
Midpoint

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(0.4,0.4)
local B = complex(-0.8,-0.3)
local M = hyper.midpoint(A,B)
hyper.drawSegment(A,B, "teal")
hyper.drawPoints(A,B,M)
hyper.labelPoints(A, "$A$", B, "$B$", M, "$M$")
\end{luacode*}
\end{hyperbolicdisk}
```



Interpolate and barycenter

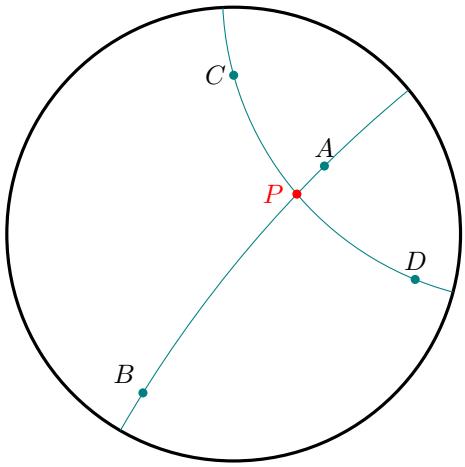
```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(0.3,0.8)
local B = complex(-0.7,-0.4)
hyper.drawSegment(A,B, "teal")
for k=0,10 do
  local M = hyper.interpolate(A, B, k/10)
  hyper.drawPoint(M)
  if k % 2 == 0 then
    hyper.labelPoint(M, "$M_{" .. k .. "}$", "
        ↪ below right")
  end
end
\end{luacode*}
\end{hyperbolicdisk}
```



3.2 Intersections

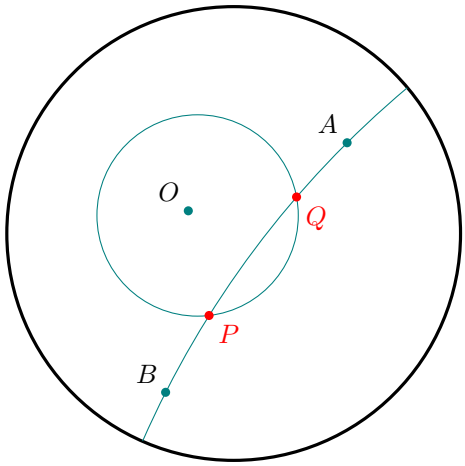
Line-Line (LL) intersection

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(0.4,0.3)
local B = complex(-0.4,-0.7)
local C = complex(0,0.7)
local D = complex(0.8,-0.2)
local P = hyper.interLL(A, B, C, D)
hyper.drawLines(A, B, C, D, "teal")
hyper.drawPoints(A, B, C, D,"teal")
hyper.drawPoint(P,"red")
hyper.labelPoint(A, "$A$", "above")
hyper.labelPoint(B, "$B$")
hyper.labelPoint(C, "$C$", "left")
hyper.labelPoint(D, "$D$", "above")
hyper.labelPoint(P, "$P$", "left=.1cm, red")
\end{luacode*}
\end{hyperbolicdisk}
```



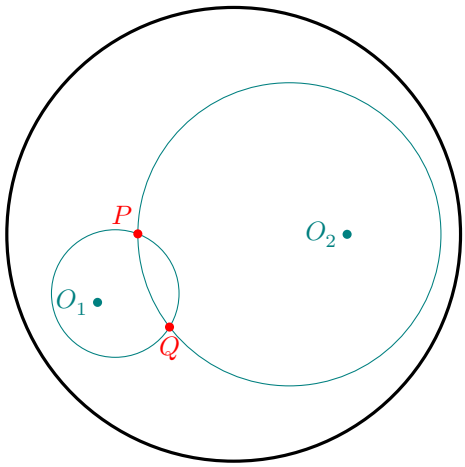
Line-circle (LC) intersection

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(0.5,0.4)
local B = complex(-0.3,-0.7)
local O = complex(-0.2,0.1)
local radius = 1
local P, Q = hyper.interLC(A, B, O, radius)
hyper.drawLine(A, B, "teal")
hyper.drawCircle(O,radius, "teal")
hyper.drawPoints(A, B, O, "teal")
hyper.drawPoints(P, Q, "red")
hyper.labelPoints(A, "$A$", B, "$B$", O, "$O$")
hyper.labelPoints(P, "$P$", Q, "$Q$", "red,
    ↪ below right")
\end{luacode*}
\end{hyperbolicdisk}
```



Circle-circle (CC) intersection

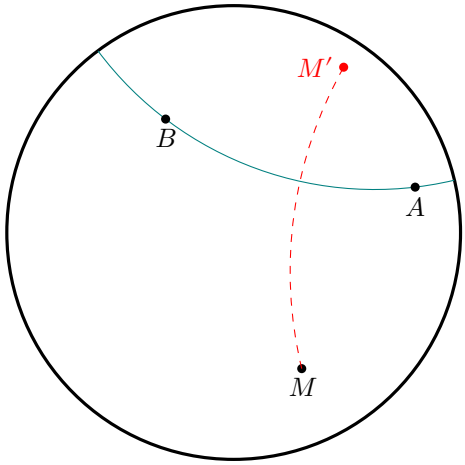
```
\begin{hyperbolicdisk}
\begin{luacode*}
local O1 = complex(-0.6,-.3)
local O2 = complex(0.5,0)
local r1, r2 = 1, 2
local P, Q = hyper.interCC(O1, r1, O2, r2)
hyper.drawCircle(O1,r1, "teal")
hyper.drawCircle(O2,r2, "teal")
hyper.drawPoints(P, Q, "red")
hyper.drawPoints(O1, O2, "teal")
hyper.labelPoints(O1, "$O_1$", O2, "$O_2$", "
    ↪ left, teal")
hyper.labelPoints(P, "$P$", "above left, red")
hyper.labelPoint(Q, "$Q$", "red, below")
\end{luacode*}
\end{hyperbolicdisk}
```



3.3 Reflections and projections

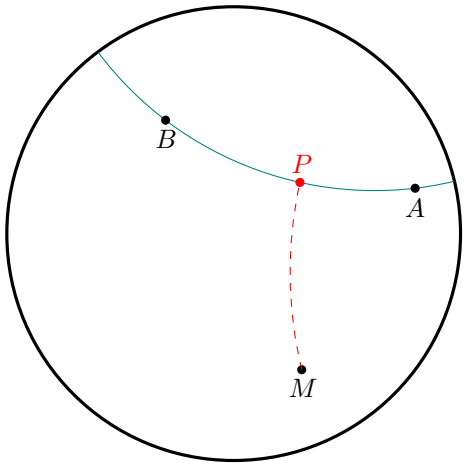
Reflection across a geodesic line

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(0.8,0.2)
local B = complex(-0.3,0.5)
local M = complex(0.3,-0.6)
local Mp = (hyper.reflection(A,B))(M)
hyper.drawLine(A, B, "teal")
hyper.drawPoints(A, B, M)
hyper.drawPoint(Mp,"red")
hyper.drawSegment(M,Mp,"dashed,red")
hyper.labelPoint(Mp,"$M'$", "left, red")
hyper.labelPoints(A, "$A$", B, "$B$", M, "$M$",
    ↪  "below")
\end{luacode*}
\end{hyperbolicdisk}
```



Projection onto a geodesic line

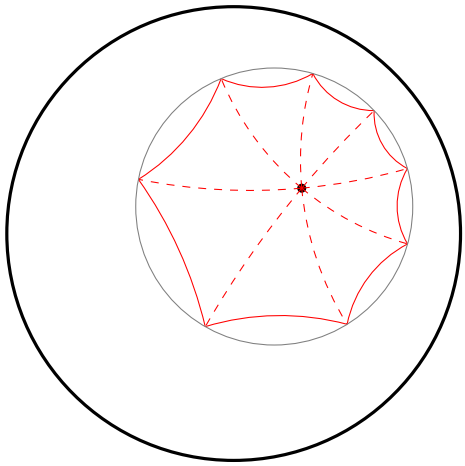
```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(0.8,0.2)
local B = complex(-0.3,0.5)
local M = complex(0.3,-0.6)
local P = (hyper.projection(A,B))(M)
hyper.drawLine(A, B, "teal")
hyper.drawPoints(A, B, M)
hyper.drawPoint(P,"red")
hyper.drawSegment(M,P,"dashed,red")
hyper.labelPoint(P,"$P$", "above, red")
hyper.labelPoints(A, "$A$", B, "$B$", M, "$M$",
    ↪  "below")
\end{luacode*}
\end{hyperbolicdisk}
```



3.4 (Elliptic) Rotations

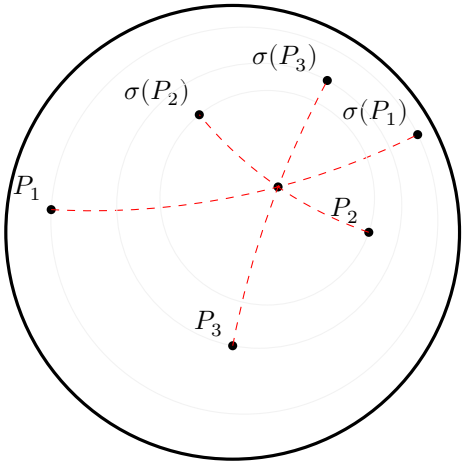
(Elliptic) rotation around a point (regular octogon)

```
\begin{hyperbolicdisk}
\begin{luacode*}
local O = complex(0.3,0.2)
local M = complex(0.5,-0.4)
hyper.drawPoints(O)
for k=1,8 do
  local P = (hyper.rotation(O,2*math.pi/8))(M)
  hyper.drawSegment(M, P, "red")
  hyper.drawSegment(O,M, "dashed,red")
  M = P
end
hyper.drawCircle(O,hyper.distance(O,M),"thin,
    ↪ gray")
\end{luacode*}
\end{hyperbolicdisk}
```



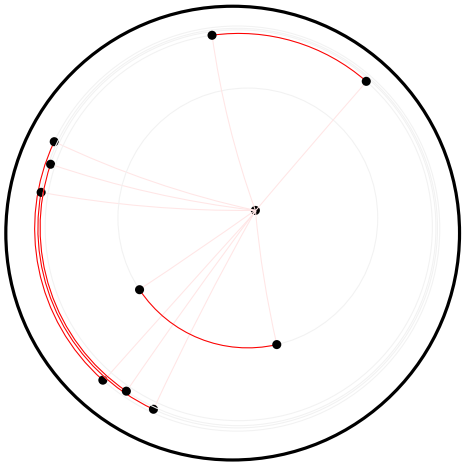
Symmetry around a point (rotation by 180°)

```
\begin{hyperbolicdisk}
\begin{luacode*}
local O = complex(0.2,0.2)
local s = hyper.symmetry(O)
hyper.drawPoints(O)
local points={complex(-.8,.1), complex(.6,0),
    ↪ complex(0,-.5)}
for k, P in ipairs(points) do
  hyper.drawCircleThrough(O, P, "gray!10")
  local sP = s(P)
  hyper.drawPoints(P,sP)
  hyper.drawSegment(P,sP, "red, dashed")
  hyper.labelPoint(P, "$P_{".. k .."}$")
  hyper.labelPoint(sP, "$\\sigma(P_{".. k .."})
      ↪ $")
end
\end{luacode*}
\end{hyperbolicdisk}
```



Rotation by 60°

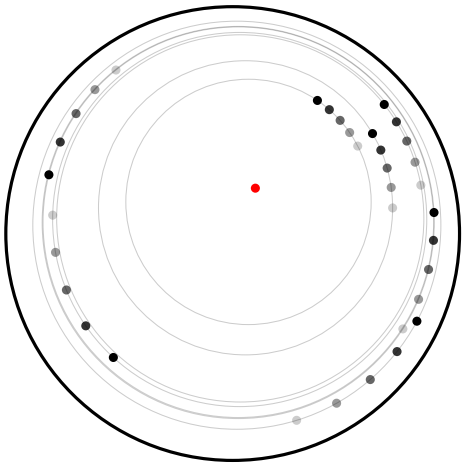
```
\begin{hyperbolicdisk}
\begin{luacode*}
local O = complex(0.1,0.1)
local r = hyper.rotation(O,math.pi/3)
hyper.drawPoints(O)
for k=1,5 do
  local P = hyper.randomPoint(0.3,0.9)
  local rP = r(P)
  hyper.drawCircleThrough(O,P, "gray!10")
  hyper.drawArc(O,P,rP,"red")
  hyper.drawPolyline(P,O,rP, "red!10")
  hyper.drawPoints(P,rP)
end
\end{luacode*}
\end{hyperbolicdisk}
```



3.5 Orbits under automorphisms

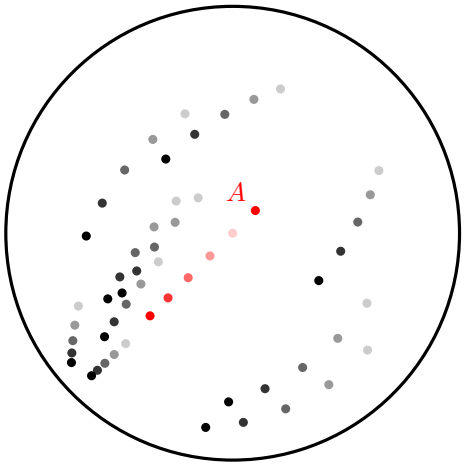
Orbit under elliptic automorphism (rotation)

```
\begin{hyperbolicdisk}
\begin{luacode*}
local nbPoints = 7
local nbIterations = 5
local O = complex(0.1,0.2)
local rot = hyper.rotation(O,math.pi/20) --
    ↪ angle in radians
hyper.drawPoint(O,"red")
for k=1,nbPoints do
  local P = hyper.randomPoint(0,.9)
  hyper.drawCircleThrough(O,P,"draw opacity=0.2
      ↪ ")
  hyper.drawPointOrbit(P, rot, nbIterations)
end
\end{luacode*}
\end{hyperbolicdisk}
```



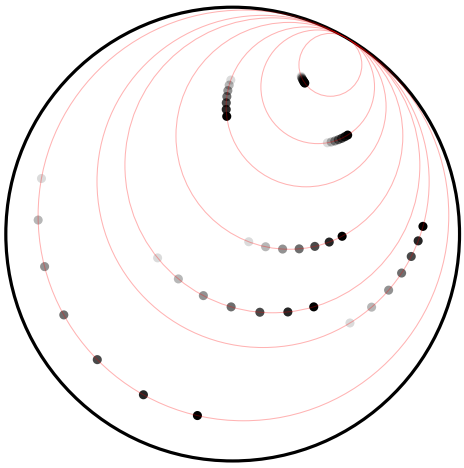
Orbit under hyperbolic automorphism

```
\begin{hyperbolicdisk}
\begin{luacode*}
local nbPoints = 10
local nbIterations = 5
local A = complex(0.1,0.1)
local phi = hyper.automorphism(A,0)
for k=1,nbPoints do
  local P = hyper.randomPoint(0,.9)
  hyper.drawPointOrbit(P, phi, nbIterations)
end
hyper.drawPoint(A,"red")
hyper.labelPoint(A,"$A$", "above left, red")
hyper.drawPointOrbit(A, phi, nbIterations, "red
    ↪ ")
\end{luacode*}
\end{hyperbolicdisk}
```



Orbit under parabolic automorphism

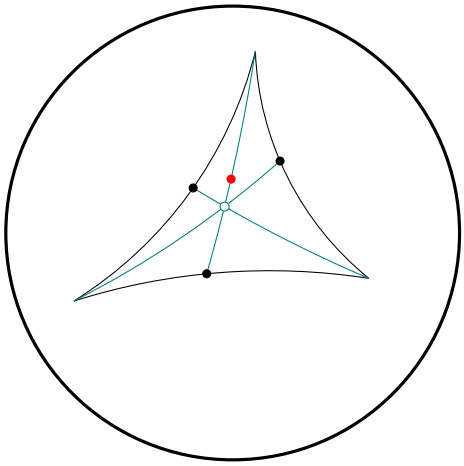
```
\begin{hyperbolicdisk}
\begin{luacode*}
local nbIterations = 7
limit = complex.exp_i(math.pi/3) -- on circle
local phi = hyper.parabolic(limit,math.pi/20)
local points={
  complex(0,0), complex(0,.7),
  complex(.4,.4), complex(.3,.7),
  complex(-.8,.4), complex(-.4,0),
  complex(.4,-.45)}
for _,P in ipairs(points) do
  hyper.drawPointOrbit(P, phi, nbIterations)
  hyper.drawHorocycle(limit, P, "red, draw
      ↪ opacity=.3")
end
\end{luacode*}
\end{hyperbolicdisk}
```



3.6 Triangle geometry (CAUTION)

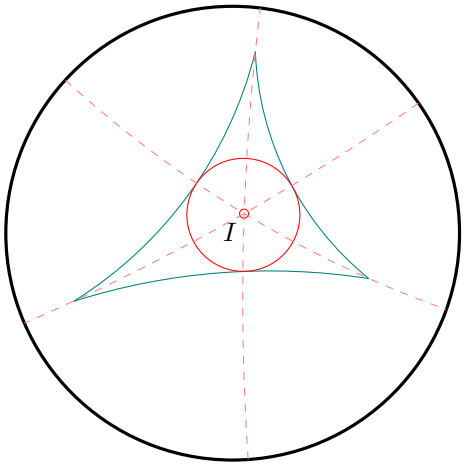
Centroid (intersection of medians)

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(0.1,0.8)
local B = complex(-0.7,-0.3)
local C = complex(0.6,-0.2)
local AA = hyper.midpoint(B,C)
local BB = hyper.midpoint(C,A)
local CC = hyper.midpoint(A,B)
local G = hyper.triangleCentroid(A,B,C)
hyper.drawTriangle(A,B,C)
hyper.drawSegments(A, AA, B, BB, C, CC, "teal")
hyper.drawPoints(AA,BB,CC)
hyper.drawPoint(G,"white, draw=teal")
-- CAUTION :
local wrong = hyper.interpolate(A,AA,2/3)
hyper.drawPoint(wrong, "red")
\end{luacode*}
\end{hyperbolicdisk}
```



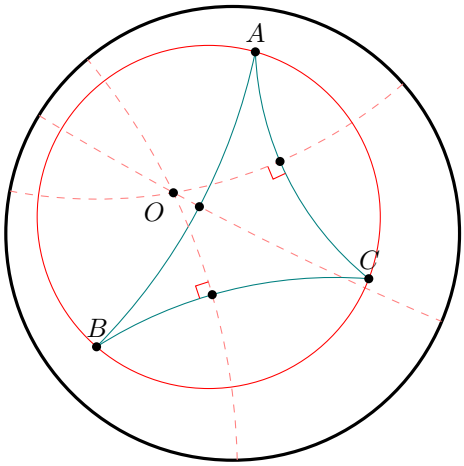
Incenter and incircle

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(0.1,0.8)
local B = complex(-0.7,-0.3)
local C = complex(0.6,-0.2)
local I = hyper.triangleIncenter(A,B,C)
hyper.drawTriangle(A,B,C,"teal")
hyper.drawPoint(I,"white, draw=red")
hyper.labelPoint(I,"$I$", "below left")
hyper.drawIncircle(A,B,C,"red")
-- show intersection :
local mystyle = "dashed, red!50"
hyper.drawAngleBisector(A,B, C, mystyle)
hyper.drawAngleBisector(B,C,A, mystyle)
hyper.drawAngleBisector(C,A,B, mystyle)
\end{luacode*}
\end{hyperbolicdisk}
```



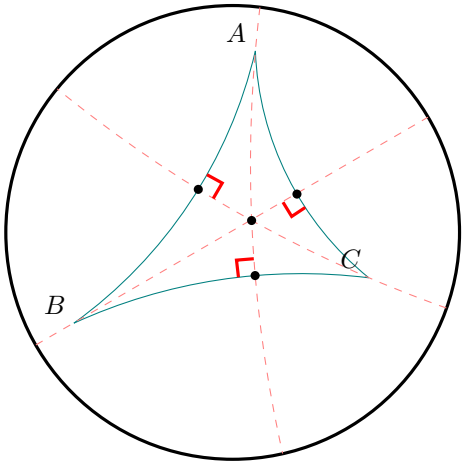
Circumcenter and circumcircle (CAUTION : does not always exist)

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(0.1,0.8)
local B = complex(-0.6,-0.5)
local C = complex(0.6,-0.2)
hyper.drawTriangle(A,B,C,"teal")
local mystyle = "dashed, red!50"
hyper.drawPerpendicularBisector(A, B, mystyle)
hyper.drawPerpendicularBisector(B, C, mystyle)
hyper.drawPerpendicularBisector(C, A, mystyle)
local O = hyper.triangleCircumcenter(A,B,C)
hyper.drawCircumcircle(A,B,C,"red")
hyper.labelPoint(O,"$O$", "below left")
local BC = hyper.midpoint(B,C)
local CA = hyper.midpoint(C,A)
local AB = hyper.midpoint(A,B)
hyper.drawRightAngle(O, CA, C,"red",1/8)
hyper.drawRightAngle(O,BC,B,"red",1/8)
hyper.drawPoints(O,A,B,C,AB,BC,CA)
hyper.labelPoints(A, "$A$", B, "$B$", C, "$C$",
    ↪ "above")
\end{luacode*}
\end{hyperbolicdisk}
```



Orthocenter (CAUTION : does not always exist)

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(0.1,0.8)
local B = complex(-0.7,-0.4)
local C = complex(0.6,-0.2)
hyper.drawTriangle(A,B,C,"teal")
local mystyle = "dashed, red!50"
hyper.drawPerpendicularThrough(A,B,C,mystyle)
hyper.drawPerpendicularThrough(B,C,A,mystyle)
hyper.drawPerpendicularThrough(C,A,B,mystyle)
local AA = hyper.projection(B,C)(A)
local BB = hyper.projection(C,A)(B)
local CC = hyper.projection(A,B)(C)
local style2 = "very thick, red"
hyper.drawRightAngle(A,AA,B,style2,1/15)
hyper.drawRightAngle(B,BB,C,style2,1/8)
hyper.drawRightAngle(C,CC,A,style2,1/12)
hyper.labelPoints(A, "$A$", B, "$B$", C, "$C$")
local H = hyper.triangleOrthocenter(A,B,C)
hyper.drawPoints(AA,BB,CC,H,"above")
\end{luacode*}
\end{hyperbolicdisk}
```

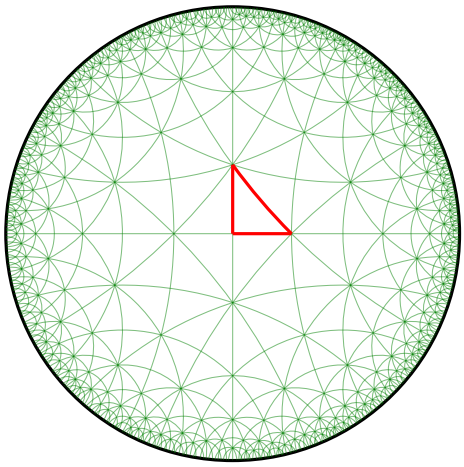


4 Tilings

All tilings in this section are draw with `TILING_DEPTH=15`

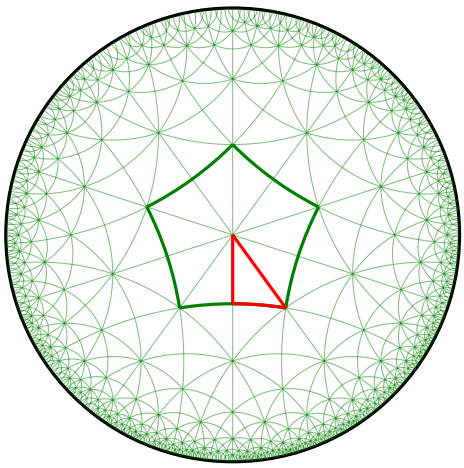
Triangle tiling

```
\begin{hyperbolicdisk}
\begin{luacode*}
hyper.SCALE=1e5
local A, B, C = hyper.fundamentalRightTriangle
    ↪ (4,5)
local geodesics = {
{hyper.endpoints(A,B)},
  {hyper.endpoints(B,C)},
  {hyper.endpoints(C,A)}}
geodesics = hyper.propagate_geodesics(geodesics
    ↪ , TILING_DEPTH)
hyper.drawLinesFromTable(geodesics, "green!50!
    ↪ black, draw opacity=.5")
hyper.drawTriangle(A,B,C,"red, very thick")
\end{luacode*}
\end{hyperbolicdisk}
```



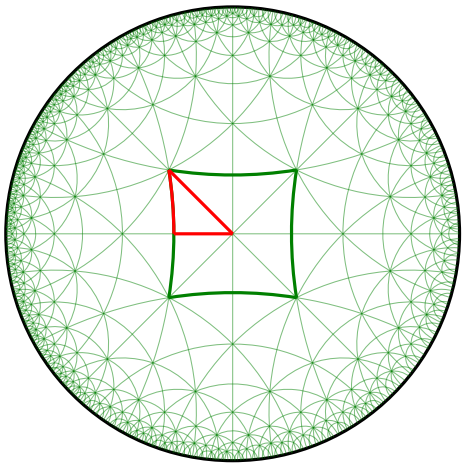
Same tiling but centered on the pentagon

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A, B, C = hyper.fundamentalRightTriangle
    ↪ (4,5)
local phi = hyper.automorphism(C,0)
A, B, C = phi(A), phi(B), phi(C)
local geodesics = {
  {hyper.endpoints(A,B)},
  {hyper.endpoints(B,C)},
  {hyper.endpoints(C,A)}}
geodesics = hyper.propagate_geodesics(geodesics
    ↪ , TILING_DEPTH)
hyper.drawLinesFromTable(geodesics, "green!50!
    ↪ black, draw opacity=.5")
hyper.drawRegularPolygon(complex(0,0),B,5, "
    ↪ very thick, green!50!black")
hyper.drawTriangle(A,B,C,"red, very thick")
\end{luacode*}
\end{hyperbolicdisk}
```



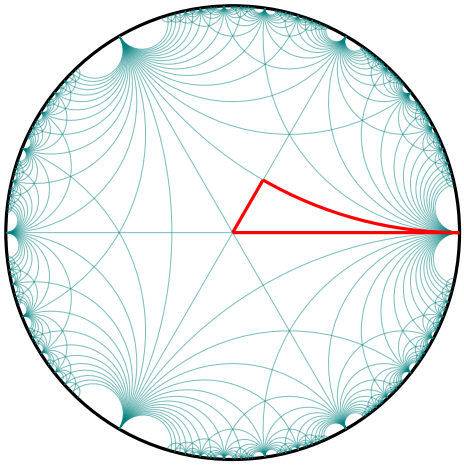
Same tiling but centered on the square

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A, B, C = hyper.fundamentalRightTriangle
    ↪ (4,5)
local phi = hyper.automorphism(B,0)
A, B, C = phi(A), phi(B), phi(C)
local geodesics = {
  {hyper.endpoints(A,B)},
  {hyper.endpoints(B,C)},
  {hyper.endpoints(C,A)}
}
geodesics = hyper.propagate_geodesics(geodesics
    ↪ , TILING_DEPTH)
hyper.drawLinesFromTable(geodesics, "green!50!
    ↪ black, draw opacity=.5")
hyper.drawRegularPolygon(complex(0,0),C,4, "
    ↪ very thick, green!50!black")
hyper.drawTriangle(A,B,C,"red, very thick")
\end{luacode*}
\end{hyperbolicdisk}
```



Ideal triangle tiling

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A, B, C = hyper.fundamentalIdealTriangle
    ↪ (3,2)
local geodesics = {
  {hyper.endpoints(A,B)},
  {hyper.endpoints(B,C)},
  {hyper.endpoints(C,A)}
}
geodesics = hyper.propagate_geodesics(geodesics
    ↪ , TILING_DEPTH)
hyper.drawLinesFromTable(geodesics, "teal, draw
    ↪  opacity=.5")
hyper.drawTriangle(A,B,C,"red, very thick")
\end{luacode*}
\end{hyperbolicdisk}
```



5 Lua libraries

5.1 complex.lua

5.1.1 Functions

__add	__unm	distinct	isColinear	isUnit	scal
__div	abs	dot	isImag	log	toPolar
__eq	abs2	exp	isInteger	new	unit
__mul	arg	exp_i	isNear	nonzero	
__pow	clone	invert	isNearInteger	polar	
__sub	conj	isClose	isNot	rotate	
__tostring	det	isClosePair	isReal	rotate90	

5.1.2 Constants (numbers and complex numbers)

EPS, EPS_INV, I, J, ONE, ZERO, __index,

5.2 luahyperbolic-tikz.lua and hyperbolic-core

5.2.1 Functions

acosh
angle
asinh
assert
assert_in_disk
atanh
automorphism
automorphism_from_pairs
automorphism_sending
barycenter2
beginTikzPicture
circle_to_euclidean
cosh
distance
distance_to_geodesic
distance_to_origin
drawAngle
drawAngleBisector
drawArc
drawCircle
drawCircleThrough
drawCircumcircle
drawHorocycle
drawIncircle
drawLabeledPoint
drawLine
drawLineFromVector
drawLines
drawLinesFromTable
drawPerpendicularBisector
drawPerpendicularThrough
drawPoint
drawPointOrbit
drawPoints
drawPolySeg
drawPolySegFromTable
drawPolygon
drawPolygonFromTable
drawPolyline
drawPolylineFromTable
drawRay
drawRayFromPoints
drawRayFromVector
drawRegularPolygon
drawRightAngle
drawSegment
drawSegments
drawSemicircle
drawTangentAt
drawTangentVector
drawTriangle
endTikzPicture
endpoints
endpoints_angle_bisector
endpoints_perpendicular_bisector
ensure_order
error
exp_map
exp_map_at_origin
fundamentalIdealTriangle
fundamentalRightTriangle
geodesic_data
in_closed_disk
in_closed_half_plane
in_disk
in_half_plane
interCC
interLC
interLL
interpolate
is_on_geodesic
labelPoint
labelPoints
metric_factor
midpoint
midpoint_to_origin
mobius_transformation
on_circle
parabolic
parabolic_fix1
projection
propagate_geodesics
radial_half
radial_scale
randomPoint
reflection
rotation
rotation_from_pair
same_geodesics
shape_closed_line
shape_euclidean_segment
shape_segment
sinh
symmetry
symmetry_around_midpoint
tangentVector
tikz_define_nodes
triangleCentroid
triangleCircumcenter
triangleIncenter
triangleOrthocenter
warning

5.2.2 Constants (numbers and complex numbers)

DRAW_EPS, DRAW_EPS_INV, DRAW_POINT_RADIUS, EPS, SCALE,